VisioVox: Audio-Visual Target-Speaker Extraction for Per-Speaker Listening in Overlapped Video, from Training to Zero-Cost Deployment

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***Abstract*—We present VisioVox, an end-to-end system that turns an uploaded multi-speaker video into one switchable audio track per visible speaker, each with word-aligned captions, played in sync with the video in a browser. On 72.7 min of AMI meetings, window-wise blind separation swaps speaker identity on 29.8% of window transitions, which motivates target-speaker extraction. We develop SEAVE, a 5.0 M-parameter TF-GridNet-style extractor with reliability-gated audio-visual conditioning, trained by curriculum. On a reverberant, overlap-verified condition its room-simulated checkpoint achieves $+8.25$ dB SI-SDR improvement, while a VoxCeleb2 fine-tune scoring $+11.43$ dB in-domain falls to -1.61 dB; blended fine-tuning recovers $+6.71$ dB. The deployed system serves a pretrained face-track-driven extractor (AV-MossFormer2) on a single workstation at zero recurring cost and a median 2.6$\times$ real time. We introduce AMI-AV, a real-recording benchmark that tiles AMI close-up cameras over summed headsets, and evaluate the deployed system on 30 meetings. The extractor gains $+3.48$ dB on overlapped speech when the face is visible, but 14.4% of speech is spoken while the speaker's face is untracked. Transcribing each separated track puts words under the right face with 110% error, because the recognizer transcribes residual cross-talk—even from close-talk headsets. Cross-track word deduplication cuts this to 78% on held-out meetings, is statistically indistinguishable from audio-only diarization in cpWER there while binding every word to a face, and, unlike single-pass attribution, keeps both speakers under heavy overlap. On the face-attributed transcript we build conversation understanding: face-anchored diarization, which names a diarizer's anonymous speakers by the separated faces, doubles overlap-detection F1 (0.38 to 0.77) and triples interruption F1 (0.07 to 0.21) at unchanged talk-share error; captions are machine-translated into 40 languages, 18 of them Indian; and locally generated summaries cite the transcript lines behind every item, finding action items with recall 0.40 and the right owner 58% of the time, but not decisions. Because each track belongs to a face, a listener can pick a person by clicking them, remix the voices, or dub each person into one of 15 languages in a synthetic voice of their own while the rest of the room is kept: on AMI-AV the original voice stays 16.5 dB down under the dub, three quarters of Hindi and Kannada lines fit as spoken, and a speech-recognition round trip recovers Hindi at median character error 0.047; the benchmark also exposed synthetic takes that were silent or ran on, now regenerated.**

***Index Terms*—target speaker extraction, audio-visual speech separation, cocktail party problem, domain shift, speaker-attributed ASR, meeting transcription, speaker diarization, conversation analysis, meeting summarization, speech synthesis, dubbing, system deployment**

I. Introduction

The cocktail-party problem [1]—following one voice among several—is easy for a listener in the room and hard for anyone listening to a recording. Overlapped speech is common in exactly the recordings people most want to revisit: meetings, interviews, panel discussions and classroom video. Existing tools address adjacent problems. Meeting transcribers attribute words to speakers but leave the audio mixed; music stem separators are not built for speech; and speech separation research [2]–[6] produces separated waveforms evaluated on fixed-length benchmark clips, not a product a person can use.

Turning separation into such a product raises three problems that benchmark numbers do not expose. *Identity over time:* a blind separator outputs sources in arbitrary order, so processing a long recording in windows and concatenating the outputs requires solving a permutation problem at every boundary. *Domain shift:* a model selected on its own validation set may behave very differently on reverberant, partially overlapped, user-recorded audio. *Integration:* a usable result needs every speaker's track aligned to the video, captions per speaker,

and a player that can switch between tracks without drift—all delivered to an ordinary browser.

This paper reports VisioVox, a working system built to address these problems, and the measurements made while building it. Our contributions are:

1) A measurement of the stitching failure of window-wise blind separation on real meetings (Section IV), which motivates target-speaker extraction (TSE) [7], where output identity is bound to a conditioning cue by construction.
2) SEAVE, a TF-GridNet-style [6] TSE model with per-frame reliability-gated audio-visual conditioning and a suppression-first objective, trained by a curriculum from anechoic to room-simulated to audio-visual data (Section V).
3) A product-condition evaluation that shows in-domain validation scores failing to predict deployed performance: the checkpoint with the highest in-domain score degrades audio on the target condition, and blended fine-tuning with min-of-domains selection recovers it (Section VII).
4) The VisioVox system: a face-track-driven AV-TSE stage, timeline placement and face- and voice-based track merging, per-speaker captioning, and synchronized multi-track web playback, deployed at zero recurring cost, with latency measured on production jobs and the integration defects that end-to-end testing exposed (Sections VI and XI).
5) AMI-AV, a real-recording audio-visual benchmark built from AMI's close-up cameras and headsets, and the first evaluation of the deployed system on it: visibility, identity and overlap-only SI-SDRi for the extractor, and speaker-attributed WER for the captions, which shows that per-track transcription fails on real meetings and that cross-track word deduplication repairs it without losing overlapped speech (Section VIII).
6) Conversation understanding on the face-attributed transcript (Section IX): an interactive transcript that plays each line in its speaker's separated voice; face-anchored diarization, which names a diarizer's turns by the separated faces and is evaluated against AMI for talk share, overlap and interruptions; caption translation into 40 languages, including Indian languages the recognizer cannot transcribe; and summaries whose every item cites transcript lines, evaluated against AMI's annotated action items and decisions.
7) Listening that the face binding makes possible (Section X): choosing a person by clicking their face, a per-speaker mixer, and per-speaker dubbing that keeps the room by subtracting only the dubbed people, evaluated on AMI-AV for leakage, timing and intelligibility; and a pre-registered user-study protocol built into the deployed system.

We report what was measured, including results that went against the original design, and we separate those measurements from claims the evidence does not yet support (Section XIII).

II. Related Work

Blind speech separation. Deep clustering [2] and permutation-invariant training (PIT) [3], [8] made speaker-independent separation trainable by scoring the best output-to-source assignment. Time-domain models such as Conv-TasNet [4], attention-based SepFormer [5], MossFormer [9], [10] and the time-frequency TF-GridNet [6] raised benchmark quality substantially. Utterance-level PIT fixes the assignment within a segment but not across independently processed segments, which is the failure we measure in Section IV.

Target-speaker extraction. TSE conditions a network on a cue identifying the desired speaker, so the output is that speaker by construction [7]. Audio cues include a d-vector (VoiceFilter [11]), adaptive layers (SpeakerBeam [12]) and jointly trained speaker encoders (SpEx+ [13]). Speaker embeddings such as ECAPA-TDNN [14] are a common enrolment representation.

Audio-visual separation. Visual cues bind a voice to a face: Looking-to-Listen [15] and The Conversation [16] condition on face or lip embeddings; AV-ConvTasNet [17], VisualVoice [18] and MuSE [19] extend this to the time domain, cross-modal consistency, and joint visual and self-enrolled cues; Michelsanti *et al.* [20] survey the field. These works are evaluated mostly on curated data where the face is frontal and visible throughout. Our deployed extractor, AV-MossFormer2 from ClearerVoice-Studio [21], conditions a MossFormer2 backbone [10] on lip crops from detected face tracks.

From models to systems. Captioning builds on large-scale ASR such as Whisper [22], and diarization toolkits such as pyannote [23] answer "who spoke when". Neither yields separated audio. Overlap in meetings is frequent and carries conversational meaning—backchannels, interruptions, competition for the floor [24]—which makes it both what a separator must handle and what a meeting reader wants to find. Multilingual translation now covers low-resource languages, including most languages of India [25], and instruction-tuned language models [26] summarise transcripts fluently but can state what their input does not; asking a model to cite its sources makes its output checkable [27]. We are not aware of published work describing a deployed system that delivers switchable per-speaker audio synchronized to video with per-speaker captions, nor of studies reporting how such a system behaves outside benchmark conditions; this paper contributes one system and those measurements, not a survey of alternatives.

III. Problem Formulation

A recording contains K speakers. With $s_k(t)$ the dry speech of speaker k, $h_k(t)$ the room impulse response to

the microphone and $n(t)$ noise, the observed single-channel mixture is

$$y(t) = \sum_{k=1}^{K} (h_k * s_k)(t) + n(t). \tag{1}$$

The video shows a subset of the speakers. For every visible speaker the system must return an estimate $\hat{s}_k(t)$ spanning the full recording, aligned sample-accurately to the video timeline, together with a word-timed transcript of $\hat{s}_k$.

Quality is measured with scale-invariant SDR [28],

$$\text{SI-SDR}(\hat{s}, s) = 10 \log_{10} \frac{\|\alpha s\|^2}{\|\alpha s - \hat{s}\|^2}, \quad \alpha = \frac{\hat{s}^\top s}{\|s\|^2}, \tag{2}$$

on zero-mean signals, reported as the improvement over returning the mixture unchanged, SI-SDRi $=$ SI-SDR$(\hat{s}, s)$ $-$ SI-SDR(y, s). A negative SI-SDRi therefore means the system made the audio worse than doing nothing.

IV. Preliminary Study: Why Not Blind Separation

We first tested the simplest design: run a pretrained two-source blind separator (SepFormer, WHAMR!-trained, 16 kHz [5]) over 4 s windows with 2 s hop and concatenate the outputs. Material was three AMI meetings [29] from three recording sites (ES2002a, IS1000a, TS3003a; 72.7 min in total), whose per-participant headset microphones provide reference signals; the mixture is their sum.

Before trusting any number, the evaluation harness was validated against a published result: SepFormer on Libri2Mix [30] (8 kHz, *min*, *mix_clean*, all 3000 test items) measured 20.58 dB SI-SDRi (95% CI [20.48, 20.67]) against the published 20.60 dB.

For each window, outputs are matched to references by SI-SDR with the Hungarian algorithm [31]; the *permutation rate* is the fraction of consecutive scored windows whose mapping changes. Windows without two simultaneously active speakers carry no ordering information and are skipped. Over 1123 scored windows the mapping changed on 334 transitions, a pooled rate of **29.8%** (per meeting 23.9–34.7%, Fig. 1a). Naively stitched tracks scored -10.70 dB SI-SDRi on average, while an oracle that assigns and rescales each window with the reference reached $+2.55$ dB from the *same* separator output (Fig. 1b). Most of the 13.25 dB gap came from per-window scale and polarity, which SI-SDR-based matching cannot see but overlap-add exposes.

The separator is not the bottleneck; stitching is. A TSE model avoids both identity and gain continuity problems because every window is conditioned on the same speaker cue and there is one output per speaker. This result fixed the architecture: *extract one speaker at a time, conditioned on a persistent identity cue*.

A second finding shaped the visual front end. A placeholder active-speaker detector based on lip-region motion energy, evaluated on AMI close-up video against headset activity, showed no diagonal structure in its face-to-voice agreement matrix (largest value 0.080). In the process we found that AMI's camera-to-headset mapping cannot be assumed: only 53 of about 171 meetings follow the naive `Closeup`$_{n+1}$/`Headset`$_n$ pairing, and our builder now reads the published mapping from the corpus metadata.

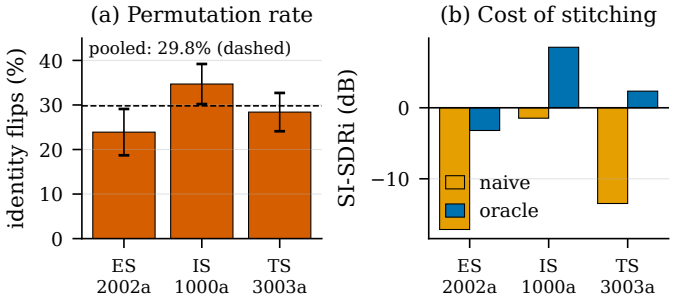


Fig. 1. Window-wise blind separation on AMI. (a) Fraction of window transitions whose output-to-speaker mapping changes (95% CI). (b) SI-SDRi of naive stitching versus an oracle assignment of the identical separator output.

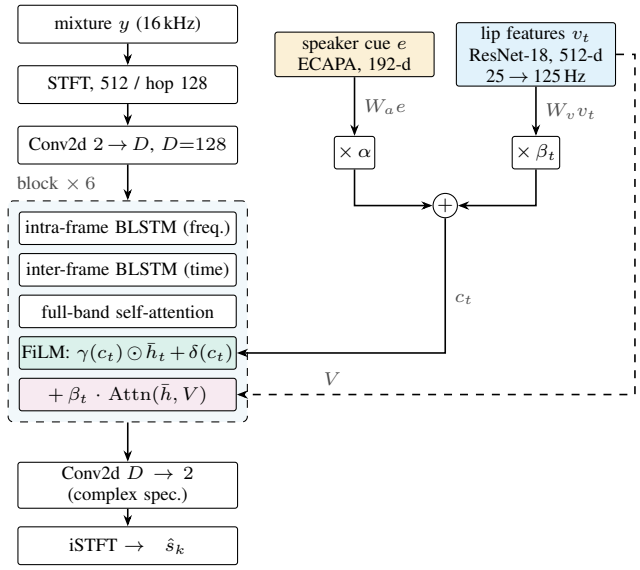


Fig. 2. SEAVE. Six TF-GridNet-style blocks map the mixture spectrogram directly to the target's. A conditioning vector c_t mixes a speaker embedding and lip features through learned reliability gates α (per utterance) and β_t (per frame) and modulates every block by FiLM; β_t also scales a cross-attention from audio to visual features V.

V. SEAVE: A Reliability-Gated Extractor

A. Architecture

SEAVE (Fig. 2) follows TF-GridNet [6]. The mixture is transformed by an STFT (512-point Hann window, hop 128, 16 kHz; 257 bins at 125 frames/s), real and imaginary parts are stacked, and a 3×3 convolution embeds them into $H \in \mathbb{R}^{D \times T \times F}$ with $D = 128$. Each of six blocks applies, with residual connections,

$$H \leftarrow H + P_f\big(\text{BLSTM}_f(\text{GN}(H))\big), \tag{3}$$

$$H \leftarrow H + P_t\big(\text{BLSTM}_t(\text{GN}(H))\big), \tag{4}$$

$$H \leftarrow H + \text{MHSA}\big(\text{LN}(\bar{H})\big) \otimes \mathbf{1}_F, \tag{5}$$

where BLSTM_f runs across frequency at each frame, BLSTM_t across time at each frequency [32], $\bar{H}$ $=$

$\frac{1}{F}\sum_f H_{:,:,f}$ is the frequency-pooled sequence and MHSA is four-head self-attention [33]. The inter-frame path is the reason for the backbone: speaker identity is a long-horizon property, and a model without such a path must re-decide whom it follows at every frame.

B. Reliability-gated conditioning

Real uploads contain head turns, occlusions and off-screen speakers; a visual cue that is wrong for two seconds should be ignored for those two seconds only. Given a speaker embedding $e \in \mathbb{R}^{192}$ with confidence ρ_a and per-frame visual features $v_t \in \mathbb{R}^{512}$ with confidences $\rho_{v,t}$,

$$\alpha = \sigma\big(g_a(\rho_a)\big), \qquad \beta_t = \sigma\big(g_v(\rho_{v,t})\big), \tag{6}$$

$$c_t = \alpha\, W_a e + \beta_t\, W_v v_t + b, \tag{7}$$

where g_a, g_v are small learned maps (1–16–1, SiLU) and $c_t \in \mathbb{R}^{256}$. After each block the pooled features are modulated by FiLM [34], $m_t = \gamma(c_t) \odot \bar{h}_t + \delta(c_t)$, initialised to the identity; when video is present, $m_t \leftarrow m_t + \beta_t\,\mathrm{Attn}(\mathrm{LN}(m_t), V, V)$, so the gate scales the attention *output* and removes an unreliable visual contribution rather than merely redistributing attention. The change $m_t - \bar{h}_t$ is broadcast back over frequency. Conditioning at every block rather than once at the input counters dilution of the speaker request through depth. A decoder convolution produces the complex spectrogram of the target directly (no mask), followed by an inverse STFT; a small head predicts an utterance-level confidence. The model has 5.02 M parameters; the visual front end (a 3D convolution over five frames followed by a ResNet-18 [35] trunk on 96×96 grayscale lip crops, 512-d at 25 fps) adds 11.46 M.

During audio-visual training, modality dropout zeroes or corrupts each cue per item (visual drop 0.20, audio-cue drop 0.15, visual corruption 0.25, audio-cue corruption 0.15) and lowers its confidence accordingly, so that the gates learn to recognise an unreliable cue rather than being told.

C. Suppression-first objective

SI-SDR penalises all error equally, but residual interference (another person talking) is far more objectionable to a listener than mild artefacts. The loss combines

$$\mathcal{L} = \mathcal{L}_{\text{SI-SDR}} + 0.5\,\mathcal{L}_{\text{sup}} + 0.2\,\mathcal{L}_{\text{cons}} + 0.3\,\mathcal{L}_{\text{MR}} + 0.5\,\mathcal{L}_{\text{sil}}, \tag{8}$$

with $\mathcal{L}_{\text{SI-SDR}} = -\text{SI-SDR}(\hat{s}, s)$; a hinged interference penalty $\mathcal{L}_{\text{sup}} = \frac{1}{J}\sum_j \max(0, \tau + \text{SI-SDR}(\hat{s}, s_j))$ over interferers s_j with $\tau = 20$ dB, which stops pushing once leakage is τ dB below the output so that it cannot be satisfied by distorting the target; a multi-resolution STFT term [36] (FFT sizes 512, 1024, 2048; spectral convergence plus log-magnitude ℓ_1); and a silence term

$$\mathcal{L}_{\text{sil}} = \frac{\mathbb{E}_{t\notin\mathcal{A}}[\,\|\hat{s}_t\|^2]}{\mathbb{E}_{t\in\mathcal{A}}[\,\|\hat{s}_t\|^2]}, \tag{9}$$

the output energy in frames where the target is inactive relative to frames where it is active ($\mathcal{A}$), which targets the 50–70% of a conversation in which each speaker is silent and leakage is unmasked. The speaker-consistency term $\mathcal{L}_{\text{cons}}$ (cosine distance between embeddings of output and enrolment) is implemented but was inactive in every run reported here, as the training logs show it at zero; we therefore make no claim about it.

TABLE I
TRAINING STAGES (VALUES READ FROM THE CHECKPOINTS AND RUN LOGS)

Stage	Data	Init.	Steps	LR	Best val.
C1	Libri2Mix + WHAM!	—	60k	1.5×10^{-4}	8.34 dB [a]
C3	+ room simulation	C1	30k	8×10^{-5}	10.92 dB [b]
C2 v2	VoxCeleb2 AV	C3	12k	6×10^{-5}	11.43 dB [c]

[a] Libri2Mix dev. [b] Room-simulated Libri2Mix dev. [c] VoxCeleb2 held-out speakers, audio-visual; 11.02 dB with only the audio cue and 11.02 dB with only the visual cue at the same step.

D. Curriculum

Training proceeded in stages, each initialised from the last (Table I). C1 learns separation on Libri2Mix [30] (16 kHz, *min*, train-100 and train-360 of LibriSpeech [37]) with WHAM! noise [38] introduced partway through. C3 keeps the speakers and objective but builds realistic mixtures: image-source [39] room responses via pyroomacoustics [40] (rooms $3\times3\times2.4$ to $10\times8\times3.5$ m, RT60 log-uniform in 0.15–0.8 s, reflection order 12), turn-taking rather than uniform overlap, per-speaker level spread, noise at 5–25 dB SNR, and a lossy codec round trip half the time. C2 adds the visual path on VoxCeleb2 [41] audio-visual mixtures built from the 118-speaker test split with speaker-disjoint validation. All stages use AdamW [42] (weight decay 0.01), linear warm-up for 500 steps then cosine decay, gradient clipping at 5, bf16 autocast and gradient checkpointing [43], on one NVIDIA RTX A5000 (24 GB). A dataloader benchmark sustained 295 items/s and 97.5% GPU utilisation for a realistically sized model.

VI. The VisioVox System

A. Overview and deployment

Fig. 3 shows the deployed system. Recurring cost is zero: the workstation that trained the models serves them, a Cloudflare tunnel provides HTTPS without a public IP or open port, the domain is a free subdomain, and authentication uses Firebase's free tier. Free hosting offers no GPU, and without one the pipeline runs 5.4× slower (Section VII-E), so the system is served from this workstation rather than moved to hosting that would have to be rented. Uploads go from the browser directly to the S3-compatible object store via multipart presigned URLs (16 MiB parts, resumable). Because SigV4 signs the `Host` header, URLs are signed for the public origin and the edge proxy forwards that host unchanged. The API never imports worker code; it sends jobs by task name, and a GPU worker with concurrency one consumes them, so a second upload queues rather than contending for GPU memory.

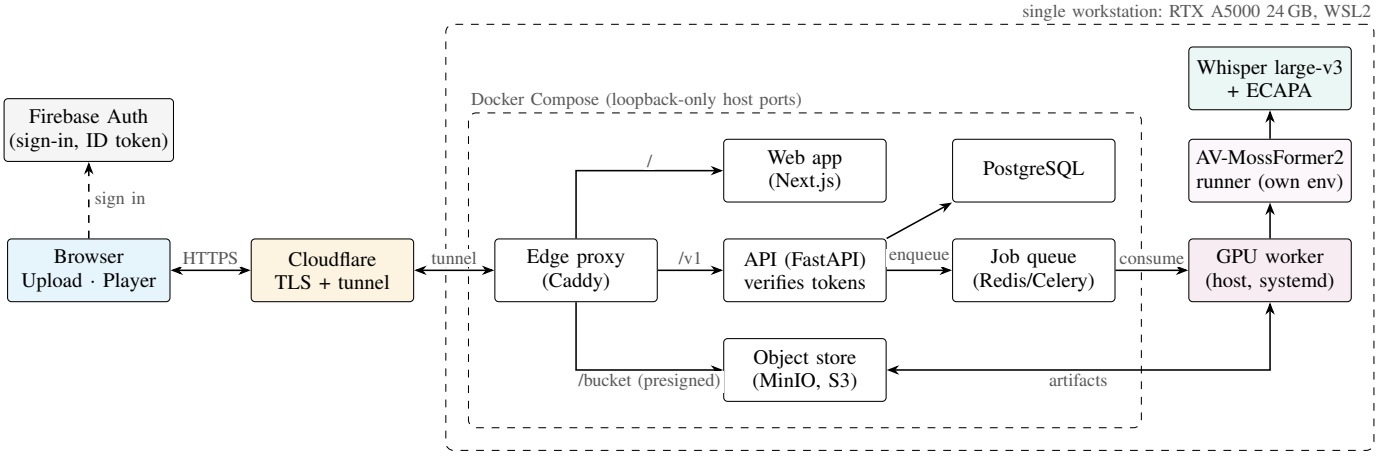


Fig. 3. Deployment. Every component runs on one workstation. The browser reaches a single public origin through a Cloudflare tunnel; an edge proxy routes `/v1` to the API, the bucket path to the object store (uploads and playback use URLs presigned for the public host) and everything else to the web app. The API verifies Firebase ID tokens and enqueues jobs by task name; the GPU worker runs on the host and reaches the data services on loopback ports.

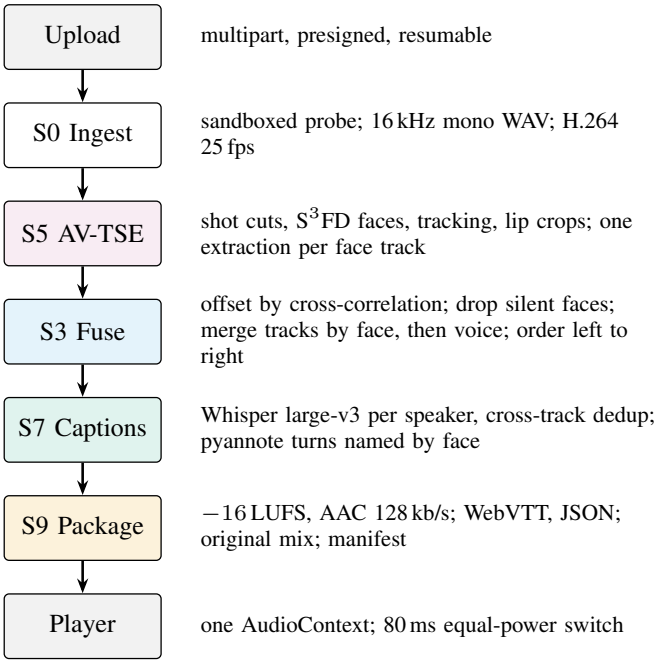


Fig. 4. Serving pipeline. A speaker is a face track: the extractor is conditioned on lip movement, and S3 turns per-track outputs into one full-length track per person.

B. Processing pipeline

The serving pipeline (Fig. 4) replaced SEAVE with AV-MossFormer2 [21] for the reasons discussed in Section XII. Its stages are:

S0 Ingest. The upload is probed in a sandbox before any decoding and normalised into a 16 kHz mono analysis WAV and a constant-25 fps H.264 video (8-bit 4:2:0 with the index first, so the same file is browser-playable). An upload without video is refused with an explanation, since the extractor requires faces.

S5 Audio-visual extraction. A separate process in the extractor's own environment detects shot boundaries, runs the S^{3}FD face detector [44] on every frame, links detections into face tracks, crops lip regions and runs AV-MossFormer2 once per track over that track's span of the analysis WAV. It returns, per track i, its frame span $[f_i^s, f_i^e)$, mean horizontal position and the separated waveform $\hat{s}_i$.

S3 Fusion: from face tracks to people. Per-track outputs cover only the frames in which a face was visible and are counted on the extractor's own re-encoded timeline. Let a be the analysis waveform and r the extractor's input waveform. The container offset is estimated, never assumed, as

$$\ell^* = \arg\max_{|\ell| \le L} \Big| \sum_t r(t+\ell)\, a(t) \Big|, \tag{10}$$

over the first 60 s with $L = 1$ s, and track i is placed at sample $640 f_i^s - \ell^*$ on a zero-initialised full-length buffer (640 samples per video frame at 16 kHz). We observed $\ell^* = 1105$ samples (69.1 ms) when the extractor's audio came from an AAC container and $\ell^* = 0$ once it was fed the analysis WAV directly. A track is discarded as a non-speaking face if

$$20 \log_{10} \frac{\mathrm{RMS}(\hat{s}_i[\mathcal{T}_i])}{\mathrm{RMS}(a[\mathcal{T}_i])} < -30\,\mathrm{dB}, \tag{11}$$

over its span $\mathcal{T}_i$. A camera cut or lost detection splits one person into several tracks, so tracks that never overlap in time are merged when their faces match—ArcFace [45] embeddings averaged over eight frames of the extractor's own crops, $\cos(f_i, f_j) > 0.3$—or, where no face can be embedded, when ECAPA [14] embeddings of their active speech satisfy $\cos(e_i, e_j) > 0.6$; tracks that overlap in time are never merged, since one face cannot be on screen twice. The face cue was added after Section VIII showed voice alone splitting people. Speakers are numbered left to right. The extractor reports no confidence, so each speaker is labelled with the fraction of the recording in which its face was tracked—an upper bound on how complete its track can be. The overlap ratio shown to the user is measured on the separated tracks as the share

of active frames (energy above -30 dB relative to the track's 95th-percentile frame) in which two or more speakers are active.

S7 Captions. Each separated track is transcribed by Whisper large-v3 [22] (faster-whisper [46], float16, Silero VAD, no conditioning on previous text) with word timestamps, so captions reflect what was recovered. Because every track also carries the other speakers a few decibels down and Whisper transcribes them, words that recur across tracks at the same moment are then kept only in the track that retains most of their energy (Section VIII). The same stage diarizes the mixture and names each diarized speaker by face (Section IX); the resulting speaking intervals ride in each speaker's caption file, so a failure here costs the conversation insights their best timing, never the job.

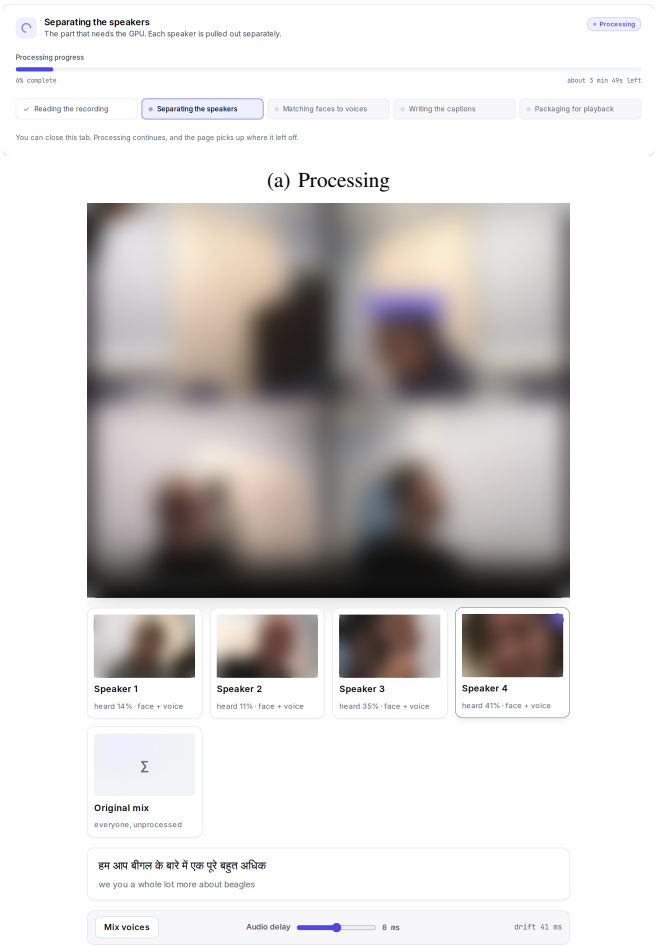
S9 Packaging and playback. Tracks are loudness-normalised to -16 LUFS (ITU-R BS.1770 [47]), encoded as AAC, and shipped with WebVTT [48] and word-level JSON captions, the unprocessed mixture as an "original mix" track, the video, and a manifest. The player decodes all tracks into one Web Audio [49] context and switches speakers with an 80 ms equal-power crossfade. Audio-video drift below 40 ms is left alone, drift between 40 and 250 ms is removed by trimming the playback rate by at most 0.4%, and larger jumps re-anchor the clocks; the active speaker's captions are highlighted word by word (Fig. 5). Decoding every track costs memory in proportion to duration times track count, so when that product exceeds 50 track-minutes (ten minutes of four speakers and the original mix) S9 also segments every track into HLS [50] renditions, and the player streams them one rendition at a time instead.

Communicating progress. Processing takes tens of seconds to minutes, so the interface names the stage the pipeline is in (reading the recording, separating the speakers, matching faces to voices, writing the captions, packaging for playback), shows its progress and the time left, and drives every element from one state model; nothing advances on a timer (Fig. 5a). The time left is learned from this server's finished jobs: a fixed and a per-second-of-video time for each stage, counted from where the job is, which predicted held-out past jobs (leave-one-out) with a median error of 16%, against 27% for a flat real-time factor. Progress arrives over server-sent events with a polling fallback, the result appears as soon as the job ends, and the page resumes where it was if closed.

VII. Experiments and Results

A. Training behaviour

Fig. 6 shows validation SI-SDRi during each stage. C1 reached 8.34 dB on Libri2Mix dev. C3, initialised from C1 and evaluated on room-simulated dev mixtures (a harder distribution), reached 10.92 dB at step 27k. On VoxCeleb2, C2 v2 reached 11.43 dB with both cues; the audio-only and visual-only paths reached 11.02 dB each, so on this data the visual cue added about 0.4 dB over the voice embedding alone.



(a) Processing

(b) Player, speaker 4 selected (faces blurred for this paper)

Fig. 5. Production interface on an AMI meeting, captured at 2× from the live site with a test account. (a) The current stage, progress, the time left as learned from this server's finished jobs, and the five steps the pipeline actually runs. (b) Per-speaker tracks with how much of the recording each person is heard in, the original mix, and the selected speaker's caption, here machine-translated into Hindi with the words as spoken beneath.

B. Product-condition evaluation

Each validation set in Fig. 6 measures something adjacent to the deployed task. Libri2Mix has overlap but no room; AMI has rooms, but measured properly contains little simultaneous speech (across ten meetings, one 4 s window had two people both clearly talking), so averaging over it scores a separator on audio that needs no separation. We therefore built the condition directly: real speech from Libri2Mix dev, simulated rooms with the C3 simulator, a requested overlap of 30–70%, and items discarded unless the overlap measured on the reverberant sources is at least 25%. Items are generated from a fixed seed, so every model sees identical mixtures (181 two-speaker and 200 three-speaker items; measured overlap 31% and 36%; mixture SI-SDR $+0.36$ and -3.99 dB). Confidence intervals are bootstrap 95% intervals [51]. The test is audio-only, so the audio-visual models are evaluated through their audio-cue path.

Table II and Fig. 7 show the result. C3 is the only model that

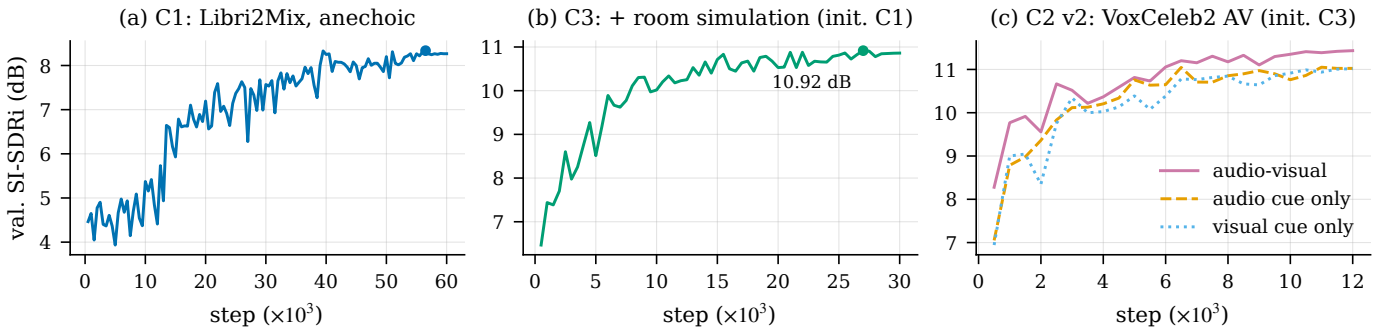


Fig. 6. Validation SI-SDRi during training. Each stage is validated on its own development data, so the panels are not on a common scale.

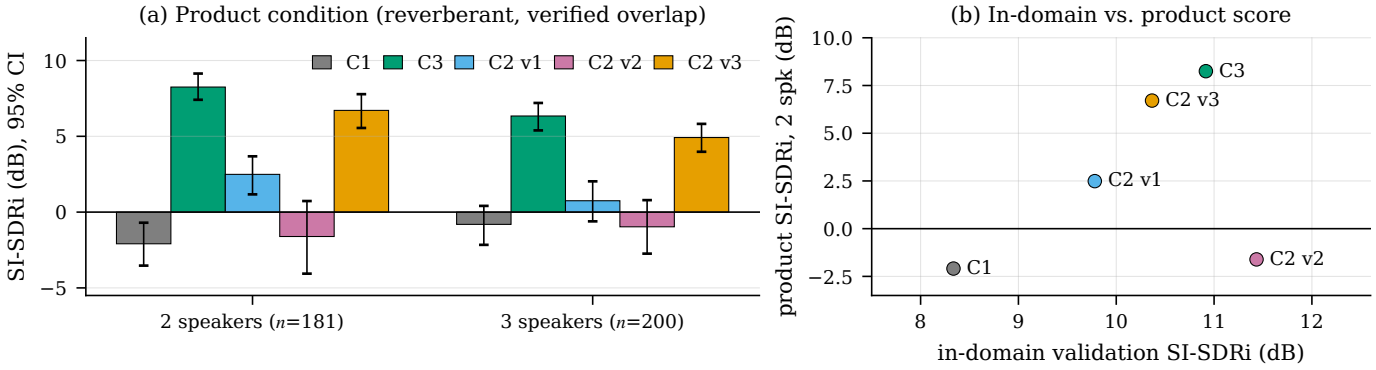


Fig. 7. (a) SI-SDRi on the product condition (bootstrap 95% CI). (b) Each model's best in-domain validation score against its two-speaker product score: the ordering is not preserved.

TABLE II
SI-SDRi (dB) ON THE PRODUCT CONDITION

		2 speakers (n=181)		3 speakers (n=200)	
Model	Data	SI-SDRi	95% CI	SI-SDRi	95% CI
C1	Libri2Mix	−2.09	[−3.53, −0.70]	−0.81	[−2.16, 0.41]
C3	+ room sim.	**+8.25**	[7.41, 9.14]	**+6.34**	[5.39, 7.20]
C2 v1	VoxCeleb2, 8k	+2.49	[1.17, 3.68]	+0.75	[−0.61, 2.03]
C2 v2	VoxCeleb2, 12k	−1.61	[−4.06, 0.73]	−0.97	[−2.74, 0.79]
C2 v3	blended, 4.5k	+6.71	[5.55, 7.78]	+4.92	[3.98, 5.82]
Mixture SI-SDR (input)		+0.36		−3.99	

holds up: +8.25 dB (95% CI [7.41, 9.14]) with two speakers and +6.34 dB with three. The checkpoint with the best in-domain score, C2 v2 (+11.43 dB on VoxCeleb2), scores −1.61 dB—it makes the audio worse than doing nothing—and the anechoic C1 scores −2.09 dB. C2 v1, trained for fewer steps on VoxCeleb2, falls in between (+2.49 dB). The pattern is a dose-response: the further a model trains on data without room acoustics, the more of the robustness learned in C3 it loses, and its in-domain score rises while it does so. Model selection on in-domain validation would have shipped the worst model. A further run, C2 v3, therefore blends room-simulated LibriMix into half of the micro-batches and selects the checkpoint maximising the minimum of its in-domain and product-validation scores (step 4.5k; 10.37 dB in-domain). It recovers most of the loss: +6.71 dB [5.55, 7.78] with two speakers and +4.92 dB with three, against −1.61 and −0.97 dB for C2 v2, though still below C3. Its final checkpoint scores higher here (+7.28 dB) but was not the one selected, so it is not reported as the result.

C. Deployed extractor on overlapped video

Because the product condition is audio-only, it cannot assess a lip-driven extractor. As a first audio-visual check we built three 6 s test videos, each placing two different VoxCeleb2 test-set speakers side by side (224×224 per face) over a mixture at 0 dB relative level with full overlap, and ran AV-MossFormer2 through the S5 runner. Table III reports the six extracted tracks after the container offset was removed: mean SI-SDRi +9.43 dB (range +5.67 to +12.57), with each output at −18 to −34 dB SI-SDR against the other speaker, i.e., the correct person was extracted every time. This is a small, favourable test and is not comparable with Table II; Section VIII evaluates the same extractor on real meetings.

D. Latency in production

Fig. 8 and Table IV report all 24 successful jobs processed by the deployed system during development and acceptance testing: seven distinct videos of 6.0–32.2 s, from 448×224 to 1920×3414, every one yielding two speakers. The median real-time factor (job time over video duration) was 2.62 (mean 3.38, range 1.81–8.17). The AV-TSE stage accounted for 72% of job time on average. High-resolution phone videos (1080p

TABLE III
AV-MOSSFORMER2 ON SYNTHETIC TWO-FACE OVERLAPPED VIDEOS (SI-SDR, DB). "LEAK" IS SI-SDR OF THE OUTPUT AGAINST THE OTHER SPEAKER.

Pair	Side	Mix	Output	SI-SDRi	Leak
P1	left	+0.06	+7.75	+7.70	−18.1
P1	right	+0.06	+5.73	+5.67	−30.4
P2	left	−0.09	+12.48	+12.57	−34.1
P2	right	−0.09	+10.78	+10.86	−31.3
P3	left	+0.17	+10.41	+10.24	−25.4
P3	right	+0.17	+9.74	+9.56	−26.8
Mean				**+9.43**	

TABLE IV
PRODUCTION JOBS: MEAN STAGE AND TOTAL TIMES

Dur. (s)	Resolution	n	S5 (s)	S7 (s)	Total (s)	RTF
6.02	448×224	5	15.1	6.8	24.0	3.99
10.80	720×1280	2	24.9	4.8	33.7	3.12
10.84	720×1280	1	24.7	2.9	31.6	2.91
10.86	1920×3414	2	64.8	11.5	87.2	8.03
14.96	848×478	1	26.7	5.8	35.7	2.39
18.45	1920×1080	2	58.1	23.9	99.2	5.38
32.23	1270×720	11	50.7	8.7	67.1	2.08
All		24				3.38

and above) ran at a real-time factor of 5.4–8.0, against 2.1–4.0 for videos of 1280×720 or below, where the highest value belongs to the shortest (6 s) clip and reflects fixed per-job cost; S5 runs face detection on every frame at the source resolution, which we expect to account for most of this difference, although its share of S5 was not measured separately. Captioning took 3–27 s per job.

E. Without a GPU

Free hosting offers no GPU, so whether the system can run without one decides where it can be served at zero cost. We processed one 52 s video of two people on screen (a televised debate, 48% of its speech overlapped) with the production pipeline twice on the same workstation, once as deployed and once with the GPU hidden from every process, and then dubbed the processed project into Hindi in each configuration (Table V). The CPU is an 18-core Xeon W-2295.

With the GPU hidden, the unmodified pipeline failed after 2 min. The extractor's library moves an index tensor to the GPU unconditionally, even though its input is on the CPU, and the caption loader requests half precision, which the recognizer's CPU back end refuses. With both worked around in the benchmark only (a no-op device move, and 8-bit Whisper on 18 threads), the CPU run reproduced the GPU run: the same three face tracks merged into the same two speakers, the same caption words, and separated tracks within 47.5 dB SI-SDR of the GPU's (lowest 39.5 dB), differences of floating-point rounding rather than of decisions. It took 5.4× as long, 7.8× real time, with about 14 of 18 cores busy throughout; a ten-minute recording would take over an hour

TABLE V
ONE 52 S TWO-SPEAKER VIDEO PROCESSED, THEN DUBBED INTO HINDI, ON THE GPU AND WITH THE GPU HIDDEN (SAME WORKSTATION; EACH RUN LOADS ITS MODELS)

	CPU only	GPU	ratio
Processing (S0–S9)	6.8 min	76.4 s	5.4×
real-time factor	7.84	1.46	
S5 faces + AV-TSE	5.8 min	54.1 s	6.4×
S7 captions + diarization	54.2 s	14.0 s	3.9×
CPU time used	5,887 s	178 s	
Dubbing into Hindi (19 lines)	3.9 min	101.1 s	2.3×
loading the voice model	10.3 s	34.0 s	0.30×
speaking the lines	3.5 min	56.8 s	3.7×
CPU time used	3,671 s	125 s	

(straight-line estimate). Separation gained most from the GPU (6.4×), captioning and diarization 3.9×.

Dubbing needed no workaround: translation and speech synthesis choose the CPU themselves. It also lost less, 2.3× overall. Speech is generated token by token in batches of eight lines, which leaves the GPU mostly idle (mean utilisation 21%), and the speech model loaded faster on the CPU than on the GPU (10 against 34 s, a fixed cost per dub whose cause we did not investigate). Sampling with a fixed seed makes repeated GPU dubs bit-identical, but the CPU draws different random numbers, so the two dubs are different takes of the same lines. Sixteen of the 19 translated lines were identical, the other three differing in word choice, and on those 16 Whisper recovered the synthesised speech with CER 11.8% (CPU) and 11.2% (GPU). The GPU therefore matters most for separation, on which every other feature depends.

VIII. REAL MEETINGS: THE AMI-AV EVALUATION

Table III tests whether the deployed extractor follows the right lips in a favourable synthetic setting. This section measures what it, and the system around it, does on real recordings.

A. Benchmark construction

AMI [29] records each participant twice—a close-up camera on the face and a close-talking headset—which makes a real multi-speaker video with per-speaker ground truth possible without simulation. For each of 30 meetings we take the 120 s window with the most simultaneous speech after the first minute, tile the four close-ups into one 2×2 video (704×576, 25 fps) using AMI's published channel-to-camera mapping, and pair it with the sum of the four headsets. Ten TNO meetings (TS3003a–TS3012a) are held out; the 20 Edinburgh and Idiap meetings are used only where a parameter must be fitted. Nothing is simulated except the tiling: participants turn away, look down, walk to the whiteboard and pass in front of other cameras.

Three corrections were needed, each measured before it was made. (i) *Who is speaking* comes from AMI's hand-corrected word alignments (±50 ms), not headset energy: some headsets sit near −55 dBFS and carry the room within 5 dB of their wearer, and an energy detector reported 47 s of speech for

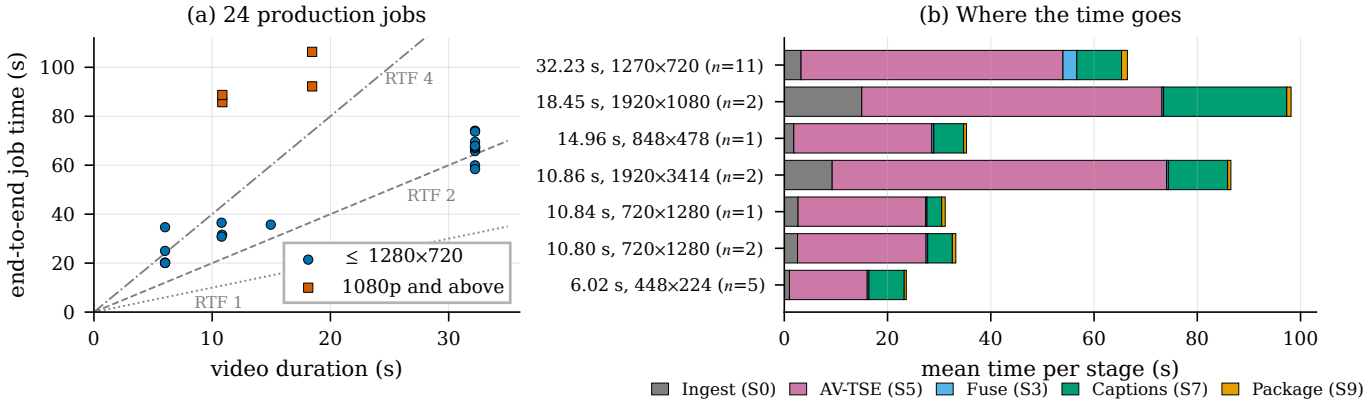


Fig. 8. Production latency. (a) End-to-end job time against video duration for 24 jobs, with real-time-factor guides. (b) Mean time per stage for each distinct video.

a participant who says nothing in the window. (ii) *Levels*: headset gains differ by up to 30 dB within a meeting, so each headset is scaled to -26 dBFS active speech before summing. (iii) *References*: a headset whose level while only others speak is within 10 dB of its wearer's speech, or whose wearer speaks for under 2 s, is not used as a signal reference (81 of 114 speakers remain). The rule was fixed before any system was scored; visibility, identity and word-error measures need no headset and use every participant. Audio-video synchrony was verified (extraction quality peaks at zero offset). Each system runs exactly as deployed, and each person it finds is assigned to a seat by where their face lies in the grid—never by which output scores best, which would hide identity errors.

B. The extractor on real meetings

Table VI summarises AV-MossFormer2 through stages S5 and S3 on all 30 meetings. Where the face is visible it helps on real overlapped speech: $+3.48$ dB SI-SDRi over the frames in which the speaker and someone else are both inside aligned words. But 14.4% of all speech is spoken while the speaker's face is not tracked (20 of 114 speakers are covered for under 90% of their speech, six never), the 120 seats yield 152 people because tracks of one person are not always merged, and when its own person is silent an output is only 3.8 dB below the mixture. Whole-clip STOI [52] falls from 0.65 to 0.54. PESQ is not reported: ITU-T P.862 is specified for 8–12 s samples, not two-minute conversations.

Merging by face. Table VI uses the original voice-only merge rule, and the split people trace to it: an extracted track carries the room, which blurs its voice print, whereas the extractor's own face crop contains nobody else. Merging time-disjoint tracks when their ArcFace [45] embeddings match (Section VI), with the threshold fitted on the 20 fitting meetings by a rule fixed in advance (fewest split people plus twice the wrong merges), turns 49 people for the 38 held-out seats that have speaking tracks (11 split, one wrongly merged) into 38, with none split and none wrongly merged. Every threshold from 0.3 to 0.6 gave no wrong merge on the fitting meetings. The deployed S3 now uses this rule.

TABLE VI
AV-MossFormer2 (S5+S3) on 30 AMI-AV meetings. SI-SDRi (dB) with 95% bootstrap intervals.

Measure	Value
Speech while face untracked	14.4% of 65 min
Speakers <90% / never covered	20 / 6 of 114
People found for 120 seats	152
Seats split into >1 person	21
People spanning two seats	3
Overlap SI-SDRi, face seen ≥90% (n=65)	$+3.48$ [$+2.61, +4.32$]
Overlap SI-SDRi, all reliable (n=71)	$+1.87$ [$-0.40, +3.54$]
Output vs. mixture, own person silent (dB)	-3.8 [$-4.6, -2.9$]
STOI, output / mixture	0.54 / 0.65

C. Who said what

Captions are what the product promises beyond audio, so they are scored against AMI's manual transcript. cpWER [53] (computed with MeetEval [54]) pairs each system's word streams with participants in whichever way minimises errors; it rewards a diarizer's anonymous but consistent clusters. VisioVox must put each caption under the right *face*, so we also report seat-attributed WER (saWER): the stream for seat k is scored against the person in seat k, with no free permutation. All systems use the deployed Whisper large-v3 (beam 5, Silero VAD on, no conditioning on previous text) and Whisper's English normaliser. Table VII gives results on the ten held-out meetings.

Per-track transcription fails, and not because of separation quality. Transcribing each person's track—the design deployed until this evaluation—places words under the right face with 110% error, most of it insertions: every track keeps the room a few decibels down, and Whisper transcribes it, so each face is captioned with most of the meeting. Transcribing each *close-talk headset* the same way is worse still (122.5% cpWER, 104.7% insertions), although bleed in those channels sits 13–25 dB below the wearer. No separator can be expected to leak less than a close-talk microphone; the remedy has to be in how words are attributed.

Cross-track deduplication. Cross-talk and genuine overlap

TABLE VII
CAPTIONS ON THE 10 HELD-OUT AMI-AV MEETINGS: POOLED WER (%) WITH 95% BOOTSTRAP INTERVALS OVER MEETINGS; INS. = INSERTIONS. †PLAIN WER, NO SPEAKERS. ‡EQUAL-LEVEL CLOSE-TALK HEADSETS.

System	saWER (%)	cpWER (%)	Ins.
Whisper, mixture†	–	31.1 [27.0, 34.5]	–
pyannote + Whisper	–	61.2 [48.0, 77.0]	12.1
Per-headset‡	–	122.5 [97.1, 147.8]	104.7
Per-track (before)	110.2 [93.1, 129.7]	96.1 [87.1, 104.4]	57.3
+ dedup (deployed)	78.1 [63.7, 94.4]	68.7 [59.9, 76.7]	16.7
+ dedup, SEAVE fill	78.3 [63.7, 94.2]	69.7 [59.9, 79.3]	19.2
Transcribe once	76.6 [64.4, 90.6]	66.1 [59.5, 73.1]	–
pyannote, by face	90.5 [77.1, 105.7]	75.2 [63.8, 88.3]	–
Oracle attribution	45.1 [39.3, 49.9]	45.1 [39.3, 49.9]	–

leave different traces in the per-track transcripts: cross-talk is the *same* word appearing in several tracks at the same moment, genuine overlap is *different* words. We therefore keep every word that appears in one track only, and for a word that recurs across tracks—identical normalised text, midpoints within 0.5 s, a rule fixed before scoring—keep only the copy in the track that retains most of the mixture's energy over it, $\arg\max_k \frac{1}{|W|}\sum_{t\in W} 10\log_{10}\bar{E}(s_k)(t)/\bar{E}(x)(t)$, with s_k track k, x the mixture, W the word's frames and $\bar{E}$ short-term energy smoothed over 50 ms (fitted on the 20 fitting meetings). On the held-out meetings this lowers saWER from 110.2% to 78.1% (paired difference -32.1 points, 95% CI $[-39.0, -24.6]$) and cpWER to 68.7%, statistically indistinguishable from pyannote 3.1 [23], [55] with Whisper on the mixture ($+7.5$ points, $[-8.8, +23.4]$). On the 20 fitting meetings pyannote is clearly better (40.4% against 63.5% cpWER), so we do not claim parity with audio-only diarization in general; the system's distinguishing property is that every word is bound to a visible face, which a diarizer alone cannot provide.

Why not transcribe once? Transcribing the mixture once and giving each word to the track that retains it is tied with deduplication on AMI ($+1.5$ points saWER, $[-1.6, +4.6]$). On the three heavily overlapped two-face videos of Table III, scored against Whisper on each clean source (a consistency check, not an absolute WER), it fails: 67.0% WER against 4.5% for per-track transcription and 5.4% with deduplication, because Whisper on the mixture transcribes one voice and the other speaker's words are lost. Deduplication is therefore what the deployed S7 now does; it was verified end to end through the public site.

What did not help. Three further designs were measured and are not deployed. *Cross-stream arbitration*—gating each track to frames where its energy share is within a margin of the largest share, with margin, floor, smoothing and hold fitted on the fitting meetings—raised frame-level speaking-activity F1 from 0.383 to 0.453 and cut per-track cpWER from 96.3% to 80.4%, but on the audio tracks it did not reduce leakage ($+0.28$ dB, $[-0.16, +1.11]$) and lowered STOI by 0.12 $[0.09, 0.14]$. *Off-screen gap filling* with SEAVE C3, enrolled from the person's own visible stretches (cosine 0.53 to an oracle voice print), raised whole-clip STOI by 0.033 $[0.013, 0.059]$ but was worse than passing the mixture in the hidden stretches themselves (-5.0 against $+3.5$ dB SI-SDR, 11 speakers) and did not improve captions (78.3% saWER). *Naming pyannote's clusters by face*, each cluster given to the track that retains most of its speech, was 12.3 points worse in saWER than deduplication $[4.6, 19.7]$. An oracle that knows who said each word of the mixture transcript reaches 45.1%: attribution, not separation, is where the remaining error lies.

IX. FROM SEPARATED TRACKS TO CONVERSATION UNDERSTANDING

A face-attributed, word-timed transcript is more than captions: it is the representation that "who said what, when" questions need, and the one that audio-only pipelines must first reconstruct by diarization. We built four uses on it—an interactive transcript, conversation insights, translated captions and grounded summaries—and measured each where a reference exists. All run on the same workstation at no recurring cost.

A. Interactive transcript

Every speaker's caption segments are merged into one conversation ordered by start time (Fig. 9a). Following playback, every line being spoken is highlighted, so during overlap two lines are active at once rather than one speaker being shown to hold the floor. Selecting a line switches the audio to that person's separated track and plays from the line's start: the transcript is a way to navigate the separation, not only to read it. Speakers can be renamed; names are stored with the project and applied to the manifest when it is served, so the player, transcript, exports and share pages read one source and the pipeline's output is never rewritten. The transcript can be searched (case- and accent-insensitive for Latin scripts; Indic scripts matched intact, since stripping combining marks changes words) and exported as SubRip, WebVTT with voice tags, plain text, or a Word document written in the browser.

B. Conversation insights

Speech is grouped into *spurts*—a person's words joined across gaps of at most 0.5 s. From spurts we report each person's talk time and share, turns, longest stretch and pace; *overlap regions*, where two or more people speak for at least 250 ms; and *overlap onsets*, when B starts at least 200 ms into A's spurt, classified following the distinction between cooperative and competitive overlap in meetings [24]: a *backchannel* if B's spurt is shorter than 1 s and ends inside A's, an *interruption* if A stops within 2 s and B outlasts A, and a *talk-over* otherwise. The definitions are implemented twice—in Python for evaluation and in TypeScript in the browser—and both are pinned to one hand-computed fixture, so the numbers users see are the numbers measured here.

Face-anchored diarization. The captions' word timings answer *who* well, because each track is bound to a face, but *when* poorly: cross-track deduplication (Section VIII) removes

TABLE VIII
CONVERSATION INSIGHTS ON 30 AMI-AV MEETINGS AGAINST THE SAME INSIGHTS FROM AMI'S WORD ALIGNMENTS. MAE OF TALK SHARE IN PERCENTAGE POINTS (95% BOOTSTRAP CI); F1 POOLED OVER MEETINGS. *LABELS NAMED BY THE REFERENCE, NOT ACHIEVABLE IN A PRODUCT.

System	Talk share MAE (pp)	Overlap F1	Interrupt. F1
Caption words of the tracks	13.1 [10.2, 16.4]	0.38	0.07
Face-anchored diarization	12.9 [9.0, 17.2]	0.77	0.21
pyannote, oracle naming*	6.4 [4.6, 8.5]	0.77	0.33

exactly the words spoken during overlap. An audio-only diarizer has the opposite profile—good boundaries, anonymous labels. We combine them. pyannote 3.1 [23], [55] segments the mixture into labelled turns; with M_ℓ the 10 ms frames of label ℓ and W_k those inside person k's caption words, labels and people are paired one-to-one by

$$\pi^* = \arg\max_{\pi} \sum_{\ell} |M_\ell \cap W_{\pi(\ell)}|, \tag{12}$$

solved with the Hungarian algorithm [31]; a label left unpaired or paired with a person whose words it never overlaps is dropped rather than guessed. Every insight is then computed from the anchored turns, with word counts (for pace) from the captions.

Table VIII scores each design against the insights computed from AMI's hand-aligned words on all 30 meetings; an interruption counts as found if the same person interrupts the same person within ± 1 s. Face anchoring doubles overlap F1 (0.38 to 0.77) and triples interruption F1 (0.07 to 0.21) at an unchanged talk-share error (-0.2 points, 95% CI $[-2.7, +2.5]$). The pairing rule was chosen on the 20 fitting meetings only: one-to-one pairing beat giving each label to its best-overlapping person (talk-share MAE 11.3 against 15.4 points) and a Jaccard-normalised variant on every measure. On the ten held-out meetings the gains in overlap (0.80 against 0.42) and interruptions (0.11 against 0.08) held, but talk share was *worse* than with caption words (16.1 against 13.1 points, n=10): when a label is paired with the wrong face, all of its talk time moves with it. The remaining gap to oracle naming ($+6.5$ points $[+3.2, +10.6]$) is that naming error, and it traces to visibility (Table VI): in TS3003a the most talkative participant's face is tracked for 11% of the window, so their words—and the label they anchor—land on other faces. Overlap F1 is identical for the anchored and oracle systems because overlap regions do not depend on naming at all.

This complements, rather than contradicts, Section VIII, where naming pyannote's clusters by face was worse for *captions*. A caption must put each word under the right face, and a mislabelled cluster moves every word in it; deduplication decides word by word. Timing measures aggregate over frames and need boundaries the per-track words cannot give. Interruption detection remains weak for every system, oracle included (F1 0.33): AMI's interruptions are short and a 2 s yield rule is coarse. The interface therefore presents interruptions and talkovers as *moments to review*, each playing the original mix from 1.5 s before it (Fig. 9b).

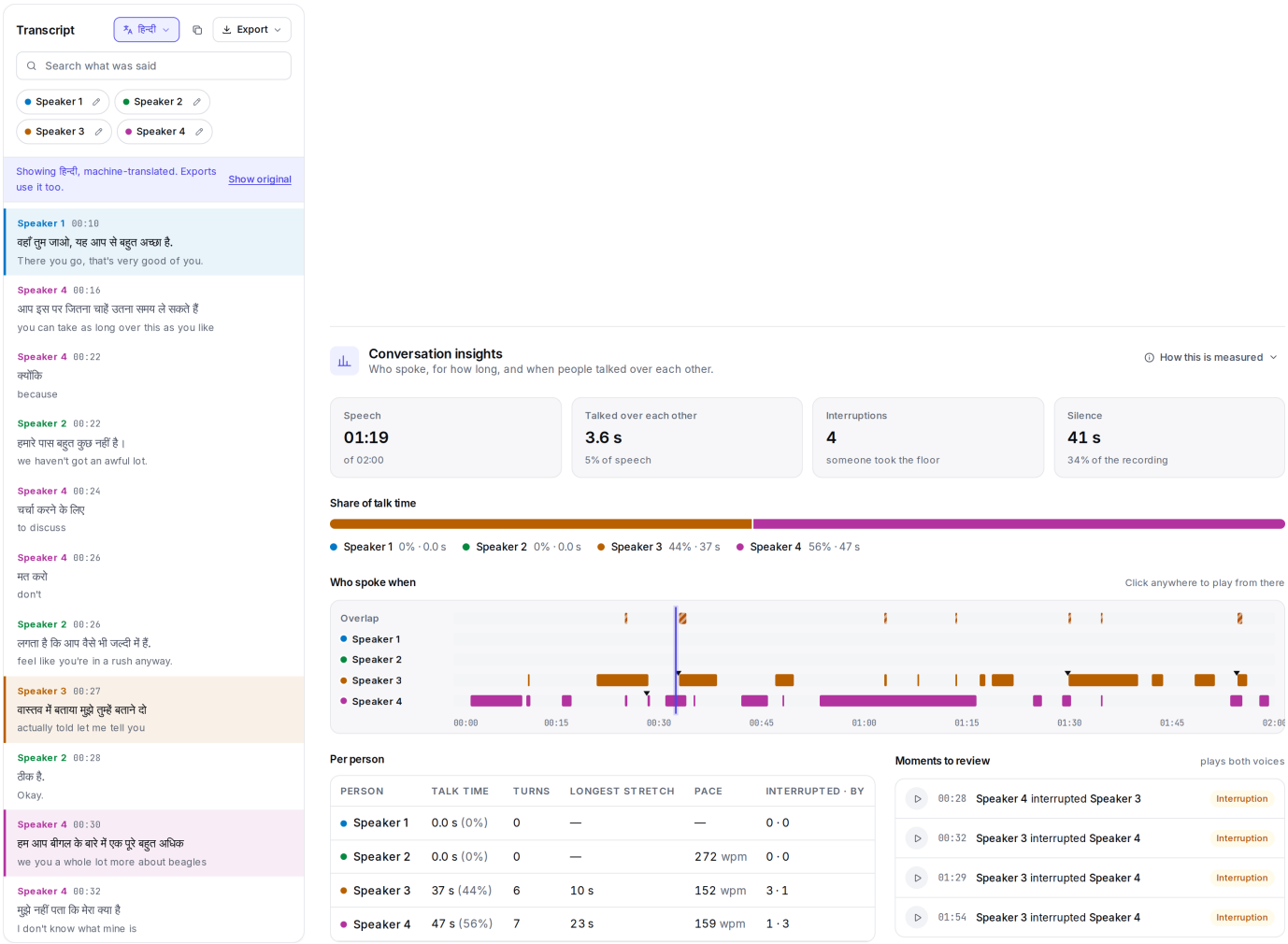
C. Translated captions

On request, every speaker's captions are translated by NLLB-200 [25] (distilled 1.3B, pretrained, used as released under its non-commercial licence) through CTranslate2 [56] with 8-bit weights, the inference engine the recognizer already uses, so translation adds a model but no runtime dependency. Each caption segment is translated as one sentence and keeps its timing; word timings are dropped, since translation reorders words. The transcript, the player's caption and every export then read in the chosen language, with the original beneath each line (Fig. 9a). Forty target languages are offered, 18 of them scheduled languages of India, including four the recognizer cannot transcribe at all—Odia, Maithili, Kashmiri and Manipuri—so a meeting held in Kannada can be read in Odia. Every candidate target was screened by translating test sentences: Santali and Konkani produced SentencePiece unknown marks and untranslated English with the released model and are not offered, and the worker rejects any translation in which more than a fifth of the segments contain unknown marks. On the workstation, 100 caption segments translate in 0.7 s once the model is loaded (4.7 s, once per worker process). Translation quality is NLLB's own [25] and was not measured on this system.

D. Summaries grounded in the transcript

A summary—overview, key points, decisions and action items with owner and deadline—is written on request by Qwen2.5-7B-Instruct [26] (4-bit) served by Ollama [57] on the same GPU; nothing leaves the machine. Language models produce fluent text that is not always supported by their input; on the first trial, the model's overview said a speaker had lost weight, which nothing in the transcript states. We therefore require attribution in the spirit of citation-generation work [27]: the prompt shows each line as `[L12 03:41 S2 Priya] text`, the answer is constrained to a JSON schema with bounded lists, and *every* item, the overview included, must list the line ids it rests on. An item is kept only if at least one cited id is a real line, and an action item's owner must be one of the speakers or none. The page shows each item with play buttons for its cited lines, so the evidence is one click away. Long transcripts are summarised in parts of 220 lines; a part the model cannot answer within its token budget is halved, down to 30 lines.

AMI's annotators recorded, for every scenario meeting, the actions agreed and the decisions taken, naming owners by role. We summarised each of the 30 meetings' *whole* transcripts as AMI annotated them, with speakers named by role, and matched written items to annotated ones lexically (content words with role names removed, Dice overlap ≥ 0.35, one-to-one), which undercounts paraphrases. This measures the summariser given a correctly attributed transcript; the separation's own errors are not compounded. Table IX gives the result. Every meeting was summarised and no item had to



(a) Transcript, read in Hindi

(b) Conversation insights

Fig. 9. Conversation understanding in the production interface (live-site capture, AMI meeting ES2002a). (a) The merged transcript follows playback—during overlap both speakers' lines are highlighted—and is shown machine-translated into Hindi with the words as spoken beneath each line; selecting a line plays that person's separated voice. (b) Talk share, a who-spoke-when timeline with overlap regions, per-person statistics and moments to review, all from face-anchored diarization.

TABLE IX
SUMMARIES OF 30 AMI MEETINGS FROM AMI'S OWN SPEAKER-ATTRIBUTED TRANSCRIPTS, AGAINST THE ACTION ITEMS AND DECISIONS AMI'S ANNOTATORS WROTE. LEXICAL MATCHING; A FLOOR.

Measure (30 meetings)	Value
Meetings summarised / failed	30 / 0
Items citing no real line (dropped)	0 of 314
Action items P / R / F1	0.29 / 0.40 / 0.34
written / annotated / matched	95 / 70 / 28
Owner right (matched, role named)	15 / 26 (58%)
Decisions P / R / F1	0.00 / 0.00 / 0.00
written / annotated	11 / 90
Median time per meeting	8.5 s

be dropped for citing nothing real. Action items were found at a useful rate (recall 0.40, precision 0.29), and the owner was right for 15 of the 26 matched items that name a role (58%; four roles, chance 25%)—for example, the model's "Review the user requirement specification", owned by the Marketing Expert, against the annotators' "The marketing expert will look at user requirement specifications" (TS3005a). Many unmatched items are activities inside the meeting, such as AMI's warm-up exercise of drawing an animal, that annotators did not record as actions. Decisions were not found: the model wrote 11 against 90 annotated, and none matched. A median meeting took 8.5 s.

E. Shareable clips with burned-in captions

People share moments, and video posted to feeds autoplays muted. An export can therefore be a clip—set by times, by the current playback position, or from transcript lines, where a second line stretches the clip—and can carry the captions in the picture, as spoken or in any translated language. Captions are written as ASS and drawn by ffmpeg's libass, which shapes Indic scripts with HarfBuzz; with several speakers each line leads with the name in that speaker's colour, and overlapping

lines stack. A single-speaker export carries that speaker's captions only, matching its audio. Cutting or drawing requires re-encoding; the whole recording without captions is still a remux. In the end-to-end test, a 12 s clip with Hindi captions was ready for download 4.3 s after it was requested.

X. Listening by Face, and Dubbing per Speaker

Because target-speaker extraction binds every output track to a face, the separated audio is not only something to read captions from. It can be acted on: a person can be chosen by pointing at them, the voices can be rebalanced, and each voice can be replaced by a translation spoken in a voice of its own while the rest of the recording stays as it was. This section describes those three uses and measures the third, which has measurable failure modes, on AMI-AV.

A. Click a face, and the listener's mix

The face tracker of S5 already follows every face frame by frame; its runner keeps each track's box every sixth frame (about four per second) as fractions of the frame, and the boxes of all tracks merged into one person (S3) travel with that person's captions. The player draws a target over each face visible at the current time (the nearest sample within 400 ms), mapped through the video's letterboxing and kept clear of the native controls. Selecting it switches the audio to that person's separated track (Fig. 10a). Pointing at a face replaces the step of matching "Speaker 2" to a person, and it is sound only because the track *is* that face's by construction.

Selecting one person is not always what a listener wants. The Web Audio engine already decodes every track onto one clock and switches by gain, so a mixer sets each track's gain from 0 to 200% with a 30 ms glide, the original mix serving as the room (Fig. 10b). The same levels can be exported: the server renders them with one gain per track, sums without renormalisation and limits peaks at -0.4 dBFS, and when captions are burned in, only the voices that are audible in the mix are shown. The streaming engine plays one rendition at a time and does not offer mixing.

B. Per-speaker dubbing

Voices. Each speaker's translated captions (Section IX) are spoken by Indic Parler-TTS [58], [59], a pretrained text-to-speech model used as released, which is conditioned on a text description of the voice and names recommended speakers for each language. Every person gets a distinct named voice, with the pitch varied when a language has fewer named voices than there are speakers, in 15 languages: 14 of India's and English. The speakers' own voices are deliberately *not* imitated, because cloning a voice from a recording its owner did not consent to is outside what the system should do. The interface states that the speech is translated and synthetic.

The room. Ordinary dubbing replaces the whole soundtrack, so laughter, other people and ambience go with it. With every person separated, only the dubbed people need removing:

$$\hat{r} = x - \sum_{k \in \mathcal{D}} g_k \, s_k, \qquad \mathbf{g} = \arg\min_{\mathbf{g}} \left\| x - \sum_{k} g_k \, s_k \right\|^2, \quad (13)$$

where x is the mixture, s_k the packaged track of person k and $\mathcal{D}$ the people with translated lines. Packaged tracks are loudness-normalised (S9), so each one's scale inside the mixture is recovered by least squares rather than assumed. A person with nothing to translate is not subtracted and stays in the original language.

TABLE X
Dubbing on 30 AMI-AV meetings, 1,467 lines per language. CER and language identification (LID) by Whisper large-v3 on the synthesised speech against the text it was given; English re-voices the original text as a control for the voice's own pace and for the recogniser.

	Hindi	Kannada	English
Lines fit as spoken	78%	74%	81%
Sped up to fit (mean tempo)	22% (1.05)	26% (1.06)	19% (1.05)
Start >0.3 s late (median lag)	4.9% (0.97 s)	6.8% (0.84 s)	3.9% (0.70 s)
Dub length / original	1.06	1.14	1.03
Median CER, all lines	0.047	0.143	0.000
Median CER, lines ≥10 words	0.059	0.108	0.040
LID, lines ≥10 words	98%	89%	99%
Synthesis time / speech	1.12	1.03	0.93

Timing. Each line starts where it was spoken. Its *slot* ends 120 ms before the same person's next line, so people may overlap each other as they did, but no one overlaps themselves. A line longer than its slot is sped up without changing pitch (by at most $1.35\times$). A line still too long pushes that person's next line later, and the lag is absorbed by the next pause. Each line is played at its speaker's original speech level, and the sum is muxed under the unchanged picture. The same run also keeps each dubbed person's new voice alone, at the level it has in the dub, and packages these tracks with the dubbed soundtrack in the shape of an ordinary project, so a dub opens in the same player as the original: one person's dubbed voice, the whole dub, or the mixer.

Takes. On one-word prompts the speech model sometimes keeps talking for seconds after the word, and sometimes produces no audio at all. A take outside $[\max(0.25, 0.04\,c),\ \max(1, 0.4{+}0.2\,c)]$ seconds for a line of c characters, or with a peak below 0.01, is generated again with other seeds. These bounds lie well outside the 80–140 ms per character of real lines. If no take is good, a long take is cut at the bound, since the word comes first and the babble after; a short take keeps its longest version.

C. Evaluation on AMI-AV

We ran the deployed code on the 30 AMI-AV meetings of Section VIII, reusing their separation and tuning no constant. Lines are AMI's hand-aligned words per participant, split at pauses of at least 0.5 s, after sentence-final punctuation, or at 25 words, with fillers dropped as the recogniser drops them. They are translated by the deployed NLLB, voiced by the deployed runner, and placed as above.

Leakage. How much of a dubbed person's original voice remains under the dub was scored against their equalised headset on the 79 reliable seats with a track; two further

reliable seats had no track, so their voice is not removed. Along each person's own headset, the room keeps their voice 16.5 dB below the mixture where they speak (13.3 dB where someone else speaks too), about 2% of its energy. The room's total level in their solo stretches falls by only 8.6 dB, because crosstalk into the other headsets and separation error remain. Fitting and held-out meetings agree (16.7 and 16.3 dB). The AMI-AV mixture has no ambience of its own, so whether the room's other sounds survive is not measured here.

Fit and intelligibility. On meetings, three quarters of the lines fit as spoken and fewer than 7% start late (Table X), because conversational pauses absorb the lag. A stand-up clip of one continuous monologue, by contrast, had 53% of its Kannada lines start late, by up to 6.5 s. Hindi needs barely more time than re-voiced English, while Kannada needs 14% more. Whisper's round trip recovers Hindi with median CER 0.047 and Kannada with 0.143. On 60-line samples, Bengali, Tamil, Telugu and Marathi gave medians of 0.13–0.19. Whisper's own errors on Indian languages are included in these numbers, so they bound the voice's errors from above; the English control (median 0.000) shows the floor when the recogniser is strong. Means are higher (Hindi 0.17, Kannada 0.29) because of one- and two-word lines, whose language Whisper names in only 19% (Hindi) and 3% (Kannada) of cases.

What the benchmark found. Its first run exposed the two take failures above: Whisper heard nothing in 63, 77 and 80 lines (Hindi, Kannada, English), most of them 0 ms long. In the product such a line had vanished from the dub after its speaker's voice had been removed from the room. With the take bounds, those counts fell to 5, 7 and 2, and mean CER fell from 0.240 to 0.174 (Hindi) and from 0.337 to 0.292 (Kannada). Table X reports the rerun. Translation quality and naturalness were not measured. The first needs references in Indian languages, and the second needs listeners.

D. Evaluation with people: protocol

None of the measurements above involves a person, so we built a user study into the deployed system and fixed its analysis before running it. It is a within-subjects comparison of VisioVox against a baseline that is the *same* player with every VisioVox feature removed, so that the controls are identical and only the features differ. Each participant uses both on two different recordings, with order and pairing counterbalanced over a four-way Latin square. On each recording they complete three tasks:

- write what one named person said during overlapped speech;
- find when a topic is mentioned;
- state an agreed action and its owner.

A facilitator page times each task from start to answer, stores participants only as codes, and exports the sessions. The three tasks are scored differently. The first counts as correct at token F1 of at least 0.5 against the reference transcript, and the second within ± 5 s. The third is scored by hand, blind to condition. The page also records the System Usability Scale [60], read against the benchmark of 68 [61], and the participant's preference.

The analysis, fixed in advance, uses a Wilcoxon signed-rank test on per-participant mean time and on accuracy, with the rank-biserial effect size and Holm correction across the two, plus a bootstrap interval for SUS and an exact binomial test for preference. Twelve to sixteen participants detect only large effects (about d=0.9 at 80% power). At the time of writing no participant had taken part, and no result is reported.

TABLE XI
DEFECTS EXPOSED ONLY BY END-TO-END TESTING

Defect	Visible symptom
API container user could not read the identity-provider key	every signed-in request failed (HTTP 500)
API imported worker packages absent from its image	uploads accepted, never processed
Presigned URLs signed for the internal storage host	uploads unreachable from browsers
Extractor fell back to CPU when `nvidia-smi` was off the service `PATH`	device-mismatch crash
Real manifest lacked the original-mix track and video	player failed to load
Caption JSON in a shape the player silently ignored	captions generated but never shown
Progress reported as a percentage, scaled by 100 again	bar full at the first update
Auth state read before session restore	"401" on opening a saved project
Concurrent first requests of a new account inserted the same user	language list empty after sign-up (HTTP 500)
Pace divided caption words by diarized talk time	a speaker at 412 words per minute
Longest turn spanned the other person's continuous speech	longest turn exceeding total talk time
Translation text kept while the next language loaded	Hindi shown under the name "Odia"

XI. LESSONS FROM END-TO-END VALIDATION

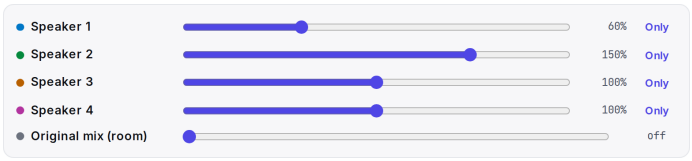
Unit and integration tests passed throughout, yet the first browser-driven end-to-end test (sign in, upload, process, play, captions) against the production stack failed in ways no lower-level test could reach. Table XI lists the defects it exposed, all now fixed and guarded by tests. They share a cause: in development every package was installed in one environment and ran as one user, and the reference pipeline's output had only ever been consumed by tests, while the player had only ever consumed mock output.

Two practices proved decisive. Measuring rather than assuming offsets and identities (Eq. (10); the AMI camera mapping) prevented silent misalignment that would otherwise have surfaced as poor quality with no visible cause. And testing the deployed artefact through a real browser found more defects in one session than all earlier testing combined.

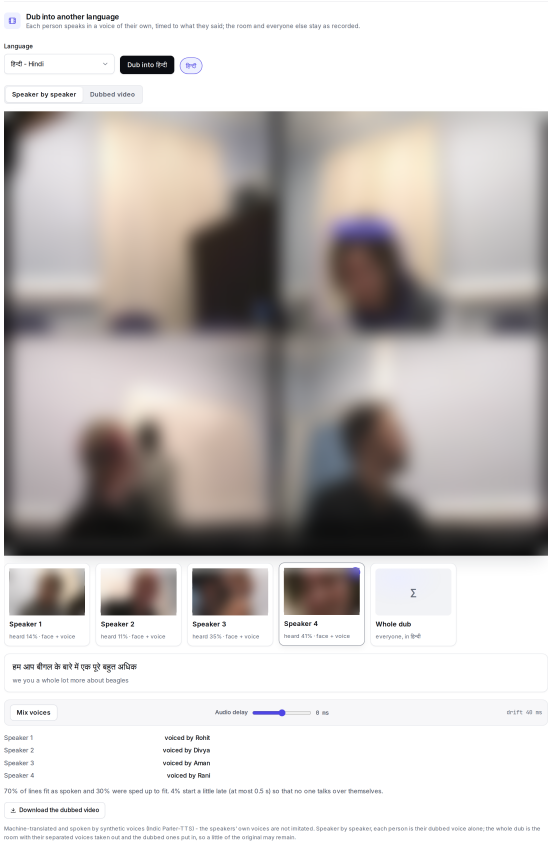
The corollary held for a path no test exercised. Video export had passed its unit tests but had never run in production, and building clips on it exposed three faults, each fatal on its own: the renderer read a source file that packaging never writes, so every video export would have degraded to audio; the rendered file was never uploaded, while the download link signed an



(a) A target over each face: the person being heard is ringed and named, another is under the pointer



(b) The mixer: each voice from off to 200%, the room separately



(c) The meeting dubbed into Hindi, speaker by speaker

Fig. 10. Listening by face and per-speaker dubbing in the production interface, on AMI meeting ES2002a (faces blurred for this paper). The dub opens in the same player as the original and reports which voice each person received and how many lines fit, were sped up or start late.

object-store key that did not exist; and the download was a plain link to an authenticated route, which a browser navigates to without the bearer token. Operations taught a last lesson. The workstation's guest file system lives in a growable virtual disk on a host drive; downloading an 11 GB model filled that drive, the guest file system went read-only, and the site was down until space was freed. On one machine every service shares fate with the disk, and free space has to be checked on the host, not inside the guest.

XII. Discussion

Why the deployed model is not SEAVE. SEAVE's audio-visual checkpoints were trained on 118 speakers and lost robustness to rooms (Table II); its best robust checkpoint, C3, is audio-only and conditions on an enrolment embedding that must first be mined from the recording by diarization. The placeholder motion-energy active-speaker detector carried no usable signal (Section IV), so binding voices to faces for SEAVE's visual path was not yet possible. AV-MossFormer2 couples face tracking with lip-conditioned extraction, removing both dependencies, and produced correct speaker identity on every synthetic test track. The switch is an engineering decision, not a claim that it is the better separator on the product condition, which remains unmeasured for it. On real meetings SEAVE found a narrower role than intended: self-enrolled from a person's visible stretches it can fill off-screen gaps, but it improved intelligibility only slightly and did not improve captions (Section VIII).

What the evaluation shows. Two results are methodological. In-domain validation inverted the ranking of models on the deployed condition, and an evaluation condition built to match the product, with overlap verified rather than assumed, was needed to see it. And on real meetings the weakest link was not separation quality but attribution: a recognizer transcribes speech far below the level any separator, or even a close-talk microphone, leaves behind, so per-speaker captions need an explicit decision about whose word each is. Cross-track deduplication is a simple such decision that is safe under overlap; the gap to an oracle (78.1% against 45.1% saWER) shows how much a learned one could still gain. The visual cue added about 0.4 dB on VoxCeleb2 for SEAVE, a thin margin that should not be generalised. Finally, the face-bound transcript and an audio-only diarizer turned out to be complements rather than competitors: words are best attributed by the faces, boundaries and overlap by the diarizer, and naming the diarizer's turns by the faces (Eq. (12)) gets both without a reference.

XIII. Limitations and Future Work

- **The deployed extractor is pretrained.** AV-MossFormer2 is used as released in ClearerVoice-Studio [21]; we did not train it. Its training data is described only as large-scale open-source data, so overlap with VoxCeleb2 cannot be ruled out for Table III. AMI is not known to be in it, but this cannot be verified either.
- **AMI-AV is one domain.** Four static close-up cameras in meeting rooms, with mostly one person speaking at a time; the 2×2 tiling places faces at fixed positions; ten meetings are held out. Heavy overlap is covered only by three synthetic videos scored against a Whisper pseudo-reference. On the fitting meetings the audio-only diarization baseline is clearly better in cpWER.
- **Product-condition speakers** come from Libri2Mix dev, which was also used, in different mixtures, to select SEAVE checkpoints; the comparison between models is fair, but absolute numbers may be optimistic.
- **No perceptual study yet.** No listening test or user study has been run; STOI is the only intelligibility measure for the separation, and caption hallucination was not measured separately from WER. The user-study protocol of Section X-D is built and fixed but has no participants yet.
- **Visibility.** Speech while a face is untracked is not recovered (14.4% of speech on AMI-AV), and every word, including one said off screen, is placed under some visible face. Audio-only uploads are refused.
- **Scale.** One GPU worker processes one job at a time, so a second upload waits. Long recordings stream over HLS, where switching speaker changes rendition and takes a few hundred milliseconds rather than 80 ms, and no mixer is offered. Without a GPU, processing takes $7.8\times$ real time (Section VII-E).
- **Data lifecycle.** Account and project deletion currently only marks records; a purge job for stored media remains to be built, which matters for a system that stores faces and voices.
- **Conversation insights** inherit the visibility limit: a label anchored to the wrong face moves talk time with it, and on the held-out meetings talk share was worse with face anchoring than with caption words (16.1 against 13.1 points, $n{=}10$). Interruptions are found with F1 0.21; they are offered as moments to review, not as counts to rely on.
- **Summaries** were evaluated on AMI's own transcripts, i.e. with perfect attribution, by lexical matching; their quality on VisioVox's transcripts is not measured, decisions were not found, and output is English only. A citation shows which lines an item is about, not that it reads them faithfully.
- **Dubbing** was measured by a recogniser, not by listeners: Whisper's errors on Indian languages are inside the reported CER, naturalness is unmeasured, and the voices are generic by design. On continuous monologue the dub falls seconds behind the picture.
- **Translation** quality was not measured here, two listed Indian languages had to be withdrawn after screening, and the translation model's licence is non-commercial—adequate for this zero-cost academic deployment, not for a commercial one.

Future work follows directly: running the user study of Section X-D; a learned active-speaker detector on the face tracks, which would both narrow the gap to oracle attribution and decide which off-screen words belong to no visible face; a reverberant audio-visual product condition with a same-input comparison against C3; summaries scored end to end on separated meetings, with a decision-extraction step the current model lacks; and translation evaluated against Indian-language references.

XIV. Conclusion

VisioVox turns a video in which people talk over each other into switchable per-speaker audio with captions, synchronized in a browser, and runs at zero recurring cost on the workstation that trained its models. Along the way we measured that window-wise blind separation scrambles identity on 29.8% of transitions in real meetings, that a reliability-gated TSE model trained with room simulation holds $+8.25$ dB on a verified-overlap reverberant condition, and that the checkpoint with the best in-domain score made audio worse on that condition. On 30 real meetings, the deployed extractor helps where it sees a face, but transcribing its per-speaker tracks captioned every face with most of the meeting; deduplicating words across tracks reduced the error from 110% to 78% on held-out meetings without losing overlapped speech. On that face-attributed transcript, naming a diarizer's turns by face doubled overlap detection and tripled interruption detection, captions became readable in 40 languages, and summaries cite the lines behind every item—each measured against AMI where a reference exists, with what did not work reported beside what did. Because every track belongs to a face, the same separation lets a listener choose a person by pointing at them and hear the meeting dubbed in Hindi or Kannada with each voice still distinct and the room intact. The broader lesson is that a separation system is validated by the condition it will serve and by the artefact users will touch, not by the benchmark on which it was trained.

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